

# Example 1: MF Ventilation Options and Split Heat Pumps

SYSTEM TAG: MF-CEXH\_ISUP-SZHP

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## PROJECT VISION

California Title 24 compliance modeling requires translating mechanical system designs into CBECC 2025 software objects, a process for which little practical guidance exists. Developed as part of the *Addressing Complex HVAC Systems in Code Compliance* project, funded by Southern California Edison through the CalBEM working group, each example walks through how a recognizable HVAC system type translates into CBECC objects, what Title 24 2025 Part 6 requires, and what assumptions were made.

### PREPARED FOR

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## Example 1: MF Ventilation Options and Split Heat Pumps

### Local Supply, Central Exhaust

System Tag: MF-CEXH\_ISUP-SZHP

#### Mechanical Description

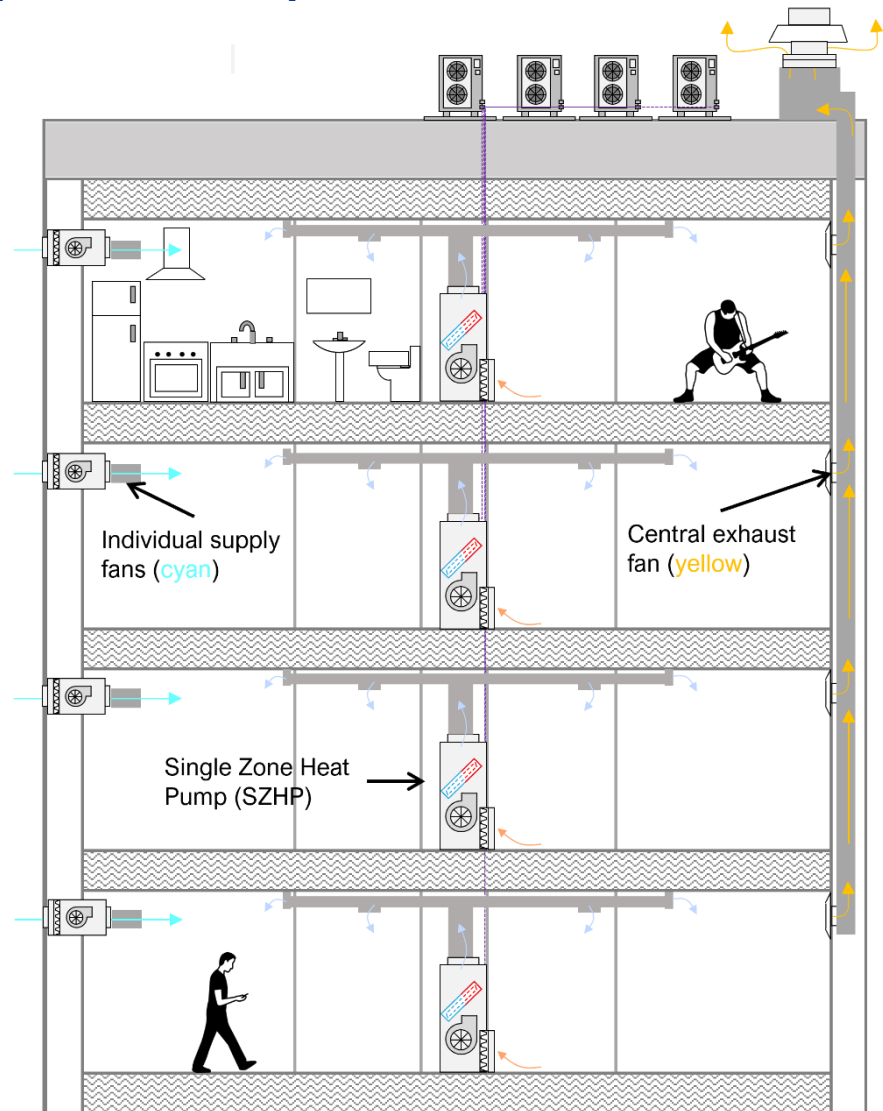
This multi-family model provides an example of ventilation configuration options in the compliance software. The option depicts one configuration here, referred to as option 1a, with additional alternate configurations in the additional example portion of this report for options 1b through 1e.

In this multi-family building, each residential dwelling unit is ventilated by an active supply fan and exhausted by a shared single exhaust fan system. The dwelling units are each conditioned by a dedicated ducted split heat pump, providing both heating and cooling. Heat pumps include electric resistance backup heat with no gas usage.

Domestic hot water is provided by a central heat pump water heater plant located in the first-floor mechanical room, with six outdoor-air-source compressors feeding a 750-gallon storage tank and a recirculation loop serving all units (not shown in diagram).

#### Diagram

The diagram shows a sample of four residential floors, each with an individual supply fan unit (left, cyan arrows) pushing outdoor air into the dwelling units, a central ducted split heat pump per floor serving the corridor zone, and a building-wide central exhaust riser (right, yellow arrows) collecting exhaust air to the roof. Outdoor condensing units are grouped on the rooftop.



## Mechanical Details

**Ventilation:** Individual supply fan per dwelling unit; building-wide central exhaust

**Ventilation control:** Fixed flow, no DCV

**Cooling:** DX cooling via heat pump, all-electric

**Heating:** Heat pump primary; electric resistance backup (no gas)

**Capacity control:** Constant volume, fixed supply air temperature

**Economizing:** None on residential/common units; Fixed Dry Bulb on retail PAVV

Ventilation is handled entirely separately from the heat pump systems through an Individual Supply / Central Exhaust (ISUP/CEXH) strategy. Each unit has a small, dedicated supply fan that draws outdoor air directly into the unit; sized by bedroom count, from 31.2 cfm for studios to 72.3 cfm for three-bedroom units.

A single building-wide central exhaust fan collects stale air from all units and expels it at the roof. Supply and exhaust airflows are balanced at approximately 4,050 cfm total. Because ventilation is driven independently by these fans, the heat pump unit fan only runs when the thermostat calls for heating or cooling, reducing fan energy during mild weather conditions.

All heat pumps share the same rated cooling and heating efficiencies, set in the equipment table below. The model uses code-minimum federal SEER2, EER2, and HSPF2 to demonstrate baseline compliance behavior. Electric resistance backup is

provided at each unit; there is no gas heating in the residential portion.

### *Mechanical Capacities and Sizing*

Residential systems are autosized per dwelling unit by the simulation engine. This was done to streamline the HVAC configuration example. The capacity range shown reflects variation in zone exposure and floor level; all units share identical input efficiency ratings.

#### Residential Ventilation Fans

| Unit Type | OA (cfm) | Supply W/cfm |
|-----------|----------|--------------|
| Studio    | 31.2     | 0.35         |
| 1-Bed     | 36.6     | 0.35         |
| 2-Bed     | 54.9     | 0.35         |
| 3-Bed     | 72.3     | 0.35         |

Central exhaust fan: 0.302 W/cfm. Supply fans are supply-only type.

#### Residential Dwelling Units

Residential dwelling unit split heat pumps were set to code minimum efficiencies for heating, cooling, and the backup electric heating device.

| Unit Type | Area (sf) | Autosized Cap. kBtu (range) * | COP | SEER2 | EER2 | HSPF2 | Backup    |
|-----------|-----------|-------------------------------|-----|-------|------|-------|-----------|
| Studio    | 540       | 6.3 – 10.7                    | 3.4 | 14.3  | 11.7 | 7.5   | Elec Res. |
| 1-Bed     | 720       | 5.5 – 39.0                    | 3.4 | 14.3  | 11.7 | 7.5   | Elec Res. |
| 2-Bed     | 1,080     | 10.6 – 31.6                   | 3.4 | 14.3  | 11.7 | 7.5   | Elec Res. |
| 3-Bed     | 1,410     | 14.4 – 22.3                   | 3.4 | 14.3  | 11.7 | 7.5   | Elec Res. |

## Code Compliance Software Configuration

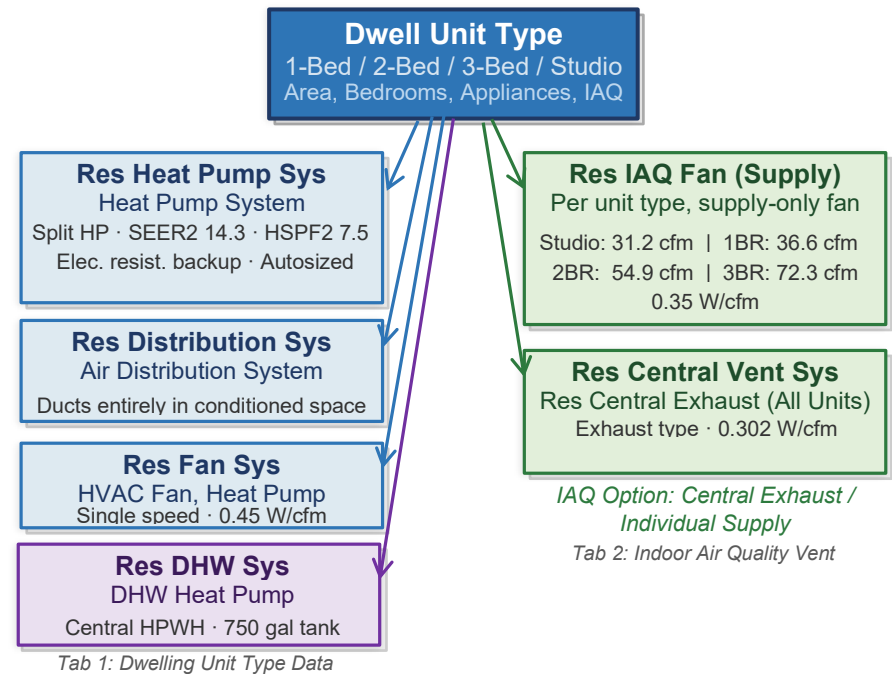
In CBECC, the residential building is configured differently from the nonresidential side. Rather than a single air system object that owns its components, each dwelling unit type is defined as a DwellUnitType, a hub object that holds references to separately defined equipment objects. This means one heat pump system specification is shared across all 88 units.

The diagram below shows how the DwellUnitType assembles its components. The left branch (blue) covers HVAC conditioning and is configured on the Dwelling Unit Type Data tab. The right branch (green) covers ventilation and is configured on the Indoor Air Quality Vent tab.

### Key Modeling Points

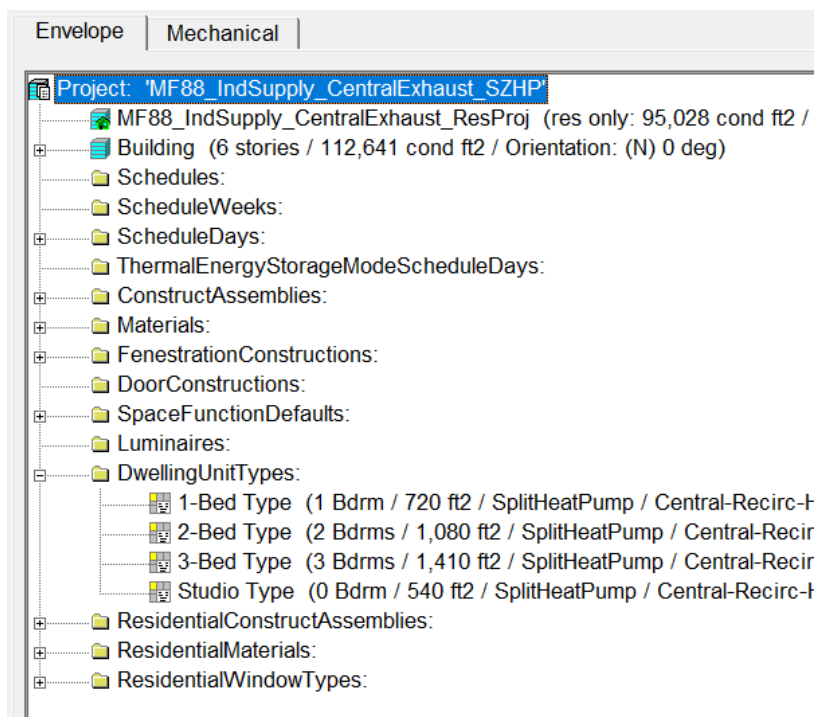
- **The DwellUnitType is a template;** it defines equipment assignments once. Individual DwellUnit instances (52 thermal zones) reference a type and inherit all assignments.
- **The Heat Pump System object** is shared across all unit types. SEER2, EER2, and HSPF2 are set once and apply to every unit; capacity is autosized per zone.
- **The IAQ approach** is set per unit type on Tab 2: Central Exhaust / Individual Supply. Each type references its own supply fan (sized by bedroom count) and a shared central exhaust system.
- **The supply IAQ fans** are set to IAQ Fan Type = Supply and do not have the option for heat recovery.

- **The central exhaust system** (Res Central Exhaust, All Units) is a single building-wide object referenced by every unit type - not duplicated per unit or floor.

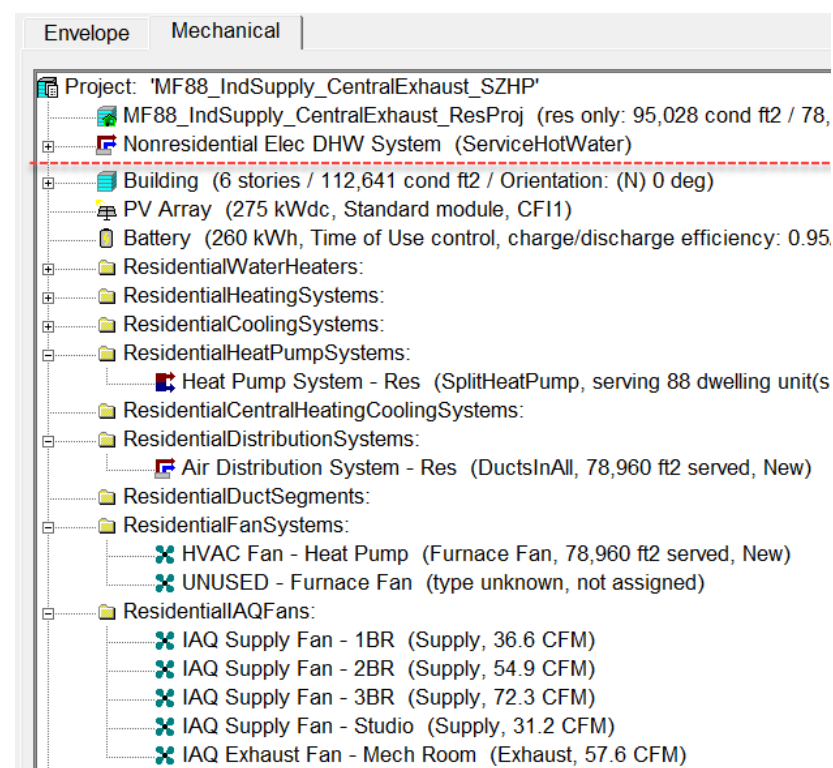


## CBECC Object Locations

For multifamily buildings in CBECC, the key objects for configuring the HVAC are on the envelope and the mechanical tab. The envelope tab, as shown below, is where dwelling units live. These are the integration points for all multifamily dwellings where all the components for each dwelling type are defined and linked to mechanical systems.

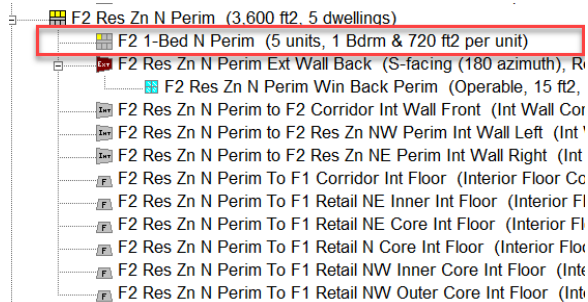


The mechanical tab is where those systems reside and where you can make multiple versions as needed of each system component. For example, this model uses a single heat pump system to represent the split heat pumps however you may end up with several of these, one for each capacity of heat pump on a project. Similarly, you can see in the residential AQ fans there are multiple fans at specific air flow rates for fresh air based on the different number of bedrooms in each typical dwelling unit.



## Thermal Space Dwelling Unit Data

Each space in the model includes a Dwelling Unit Data object. This object holds how many instances of this type are included. In this example, the space has 2 1-Bed Type units included. It is also where the washer and dryer are defined.



## Dwelling Unit Type Data (1-Bed Type shown)

The DwellUnitType is a template; it defines equipment assignments once. Individual DwellUnit instances (52 thermal zones) reference a type and inherit all assignments.

This screen sets the DwellUnit area, bedroom count, appliances, HVAC system type, heat pump object, fan, distribution, and DHW reference.

Dwelling Unit Data

Currently Active Dwelling Unit: **F2 1-Bed E Court**

Name: **F2 1-Bed E Court** 2 units, 1 Bdrm & 720 ft2 per unit

Dwelling Unit Type: **1-Bed Type** Count: **2**

Conditioned Area: **720** ft2 Area x Count: 1,440 ft2

Minimum IAQ Ventilation: 73.2 CFM

Washer Zone: **F2 Res Zn E Court**

Dryer Zone: **F2 Res Zn E Court**

Currently Active Dwelling Unit Type: **1-Bed Type**

Name: **1-Bed Type** Conditioned Area: **720** ft2 Number of Bedrooms: **1**

Appliance Data

☒ Refrigerator usage: **from # bedrooms/unit** **491** kWh/yr

☒ Dishwasher usage: **from # bedrooms/unit** **0.14507** kWh/gal

☒ Cooking appliances fuel: **Electricity**

Clothes Washer: **In-Unit**

Clothes Dryer: **In-Unit** fuel: **Electricity**

HVAC and Water Heating Equipment

HVAC System Type: **Heat Pump Heating and Cooling Sy** Distrib: **Air Distribution Sy** Fan: **HVAC Fan - Heat P**

Heat Pump(s): **1** Unique Ht Pump Unit Types **Heat Pump System - Res** Count: **1**  
1 SplitHeatPump unit(s), HSPF 8.8, auto-sized

Water Heating: **1** Unique DHW Systems **DHW Heat Pump** Fixtures, Compct Dist and Drain Wtr Ht Rec...  
1 shower(s) & 1 bath(s) served / Drain Water Heat Recov. not specified

## Indoor Air Quality Vent (1-Bed Type shown)

The IAQ approach is set per unit type on Tab 2: Central Exhaust / Individual Supply. Each type references its own supply fan (sized by bedroom count) and a shared central exhaust system.

Sets ventilation approach (Central Exhaust / Individual Supply), central vent exhaust system, in-unit supply fan, and accessibility/FID compliance checkboxes

The screenshot shows the 'Building Model Data' dialog box with the 'Indoor Air Quality Vent' tab selected. The 'Currently Active Dwelling Unit Type' is '1-Bed Type'. The 'Model as' dropdown is set to 'Central Exhaust / Individual Supply'. The 'Central Vent Sys' dropdown is set to 'Res Central Exhaust (All Units)'. The 'Minimum IAQ Ventilation' is 36.6 CFM (36.6 entered). The 'Exhaust' is 36.6 CFM. The 'Type' is 'Balanced'. The 'In-Unit Supply Fan' dropdown is set to 'IAQ Supply Fan'. The 'Cnt' is 1, and the 'CFM' is 36.6. There are two checkboxes: 'All supply air filters, outside air inlets, and H/ERV recovery cores are accessible per RACM Reference Manual' and 'IAQ system has fault indicator display (FID) in compliance with RACM Reference Manual', both of which are checked. The 'OK' button is at the bottom right.

| Field                               | Value                               |
|-------------------------------------|-------------------------------------|
| Currently Active Dwelling Unit Type | 1-Bed Type                          |
| Model as                            | Central Exhaust / Individual Supply |
| Central Vent Sys                    | Res Central Exhaust (All Units)     |
| Minimum IAQ Ventilation             | 36.6 CFM (36.6 entered)             |
| Exhaust                             | 36.6 CFM                            |
| Type                                | Balanced                            |
| In-Unit Supply Fan                  | IAQ Supply Fan                      |
| Cnt                                 | 1                                   |
| CFM                                 | 36.6                                |

☒ All supply air filters, outside air inlets, and H/ERV recovery cores are accessible per RACM Reference Manual

☒ IAQ system has fault indicator display (FID) in compliance with RACM Reference Manual

OK

## Relevant Energy Code Citations

Restructured 2025 Title 24, Part 6 with section numbers shown with legacy cross-reference in brackets.

### *Ventilation & IAQ, Attached Dwelling Units*

#### **403.2.2.3 & 403.2.2.3.4 [160.2(b)2 & 160.2(b)2Aiv]:**

The governing sections for MF dwelling unit ventilation. All attached dwelling units shall meet ASHRAE 62.2 as amended by the Code. ISUP/CEXH qualifies as balanced ventilation per item 2.1. Compartmentalization testing  $\leq 0.30$  cfm/sf at 50 Pa required per item 2.2.

### *Ducts in Conditioned Space*

#### **402.2.6.1.1, under 402.2.6.1 [150.0(m)1]:**

When all ducts are located entirely within conditioned space, duct insulation is not required. Must be confirmed through HERS field verification per RA3.1.4.3.8. All dwelling unit ducts modeled as entirely in conditioned space. No duct loss penalty applied. HERS verification required at construction.

### *Single-Zone Ducted System Fan Efficacy*

#### **402.2.6.13.2 [150.0(m)13B]:**

Single-zone ducted systems shall meet  $\leq 0.58$  W/cfm (non-gas AHU) at  $\geq 350$  cfm/ton airflow. Model inputs 0.45 W/cfm for all dwelling unit HVAC fans. HERS verification required.

### *HP Supplemental Heat Controls & Thermostat*

#### **402.2.2.7 & 402.2.4.2 [150.0(h)7 & 150.0(i)2]:**

Heat pumps with electric resistance backup must lock out supplemental heat above 35°F OAT. Thermostats must display when backup is active, include OAT sensing, and meet setback requirements.

### *Electric Resistance Supplemental Heat Sizing*

#### **402.2.2.8 [150.0(h)8]:**

The capacity of electric resistance supplemental heat shall not exceed the HP nominal cooling capacity (at 95°F)  $\times 2.7$  kW/ton, rounded up to nearest kW. CBECC autosizes the backup element. Model uses a 0.75 $\times$  heating sizing factor for the HP compressor, with backup covering the remaining load.

### *Standard Design Mechanical Package, CZ9*

#### **403.3.1.1 [170.2(c)3A]:**

For MF performance compliance in CZ9: HP SEER2 and HSPF2 at federal code minimum per 403.3.1.1; refrigerant charge verification required per 403.3.1.2.1; HRV/ERV not required for balanced ventilation. Individual supply fan efficacy for non-HRV supply systems: 0.40 W/cfm per 403.3.1.2.5 in CZ 5-10 and CZ 15.

Standard design uses same SZHP system type as proposed. Primary compliance margin comes from fan energy and lighting, not equipment efficiency.

## Alternate MF Ventilation Options

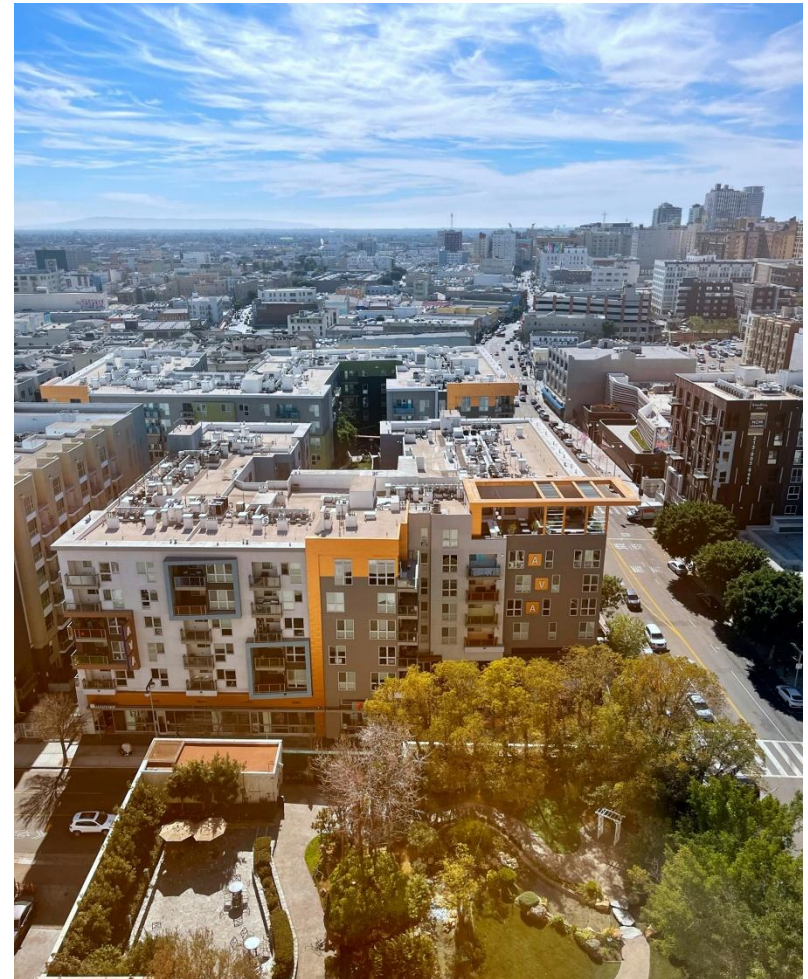
Variations of the multi-family building for other configurations of ventilation were evaluated and included as files. These include:

- a) Local Supply Fans, Central Exhaust Fans (example 1)
- b) Central Supply Fan, Local Exhaust Fans
- c) Local Supply and Exhaust Fans, Façade-Mounted
- d) Local Ventilator with Heat Recovery
- e) Central Supply Fan, Central Exhaust Fan

Examples b through e are shown here with a brief description and system depiction. Assumptions are contained in the model input file examples.

For each configuration, the fan power of the Title 24 2025 standard system is noted alongside the description, as the baseline fan power varies depending on whether ventilation is centralized or distributed at the unit level.

The baseline heat recovery ventilator assignment is also noted where relevant; configurations d and e are assigned an HRV in the standard system at SRE 0.67 / ASRE 0.72 regardless of whether the proposed design includes heat recovery, meaning a design without HRV will carry a compliance penalty against that baseline.



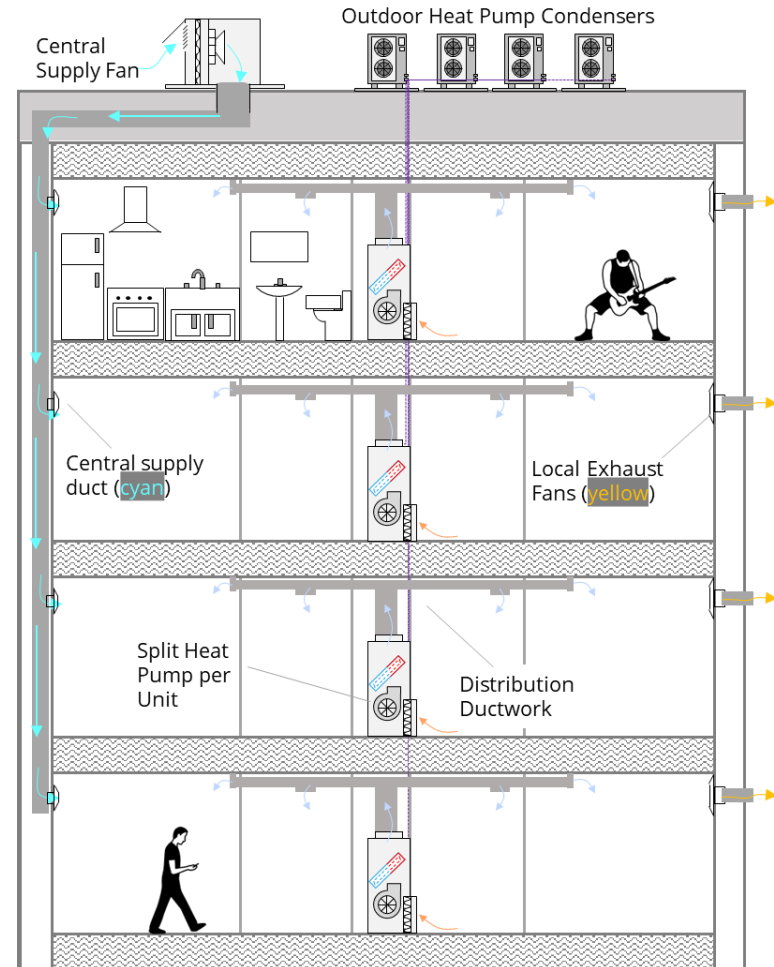
## Alternate 1b: MF Ventilation Option: Central Supply, Local Exhaust

System Tag: MF-CSUP\_IEXH-SZHP

### Mechanical Description

A single central fan delivers conditioned outdoor air via a common duct to all units. Each dwelling has its own exhaust fan (typically in the bathroom or kitchen) that operates independently. Supply and exhaust are unbalanced at the unit level. This is the most common ventilation pattern in new heat-pump multifamily construction.

The Title 24 2025 standard system for this configuration uses a central supply fan at 0.441 W/cfm and individual exhaust fans at 0.350 W/cfm, with no heat recovery ventilator in the baseline.



## Alternate 1c: MF Ventilation Option: Local Supply and Exhaust

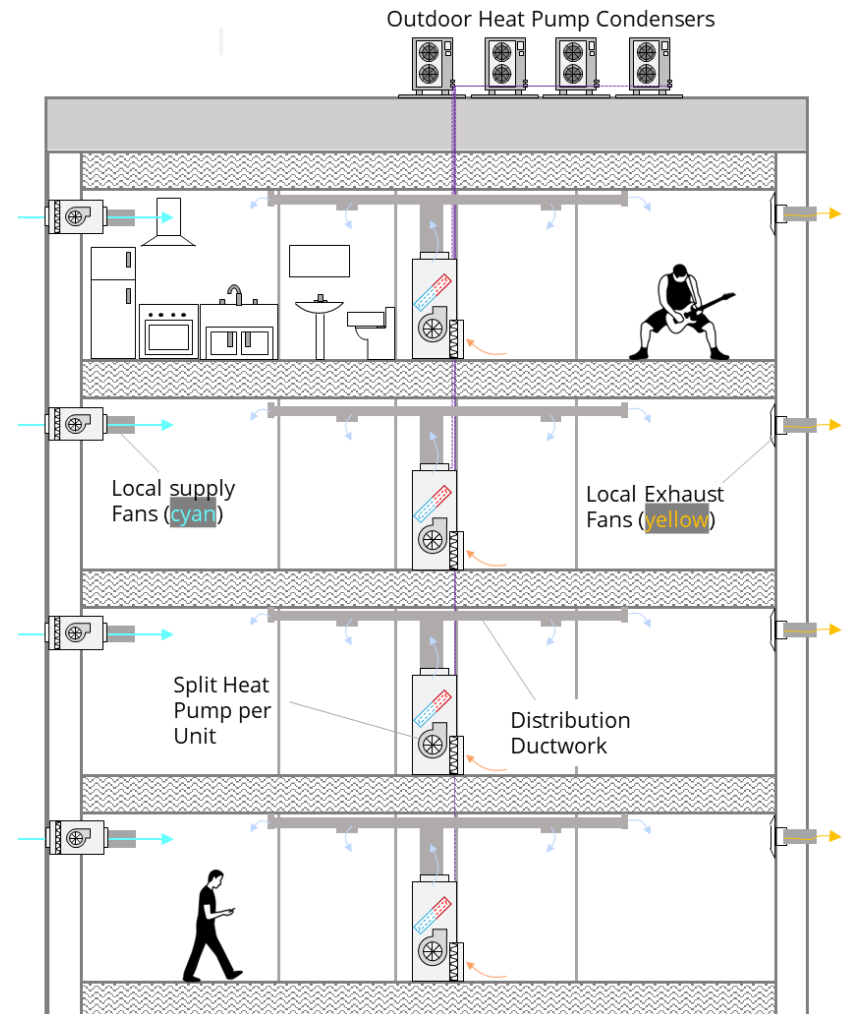
System Tag: MF-ISUP\_IEXH-SZHP

### Mechanical Description

#### **Façade-Mounted** (No Heat Recovery)

Each unit has its own supply and exhaust fan, each penetrating the exterior wall at a different location with no shared ductwork. Because the supply and exhaust streams never meet, heat recovery is not possible. This configuration is often implemented using a bathroom exhaust fan as the exhaust component, with a separate through-wall supply grille or fan serving the living area.

The Title 24 2025 standard system for this configuration is individual balanced ventilation with a combined fan power of 0.60 W/cfm and a heat recovery ventilator at SRE 0.67 / ASRE 0.72, regardless of whether the proposed design includes heat recovery. A proposed design without HRV will take a compliance penalty against this baseline.



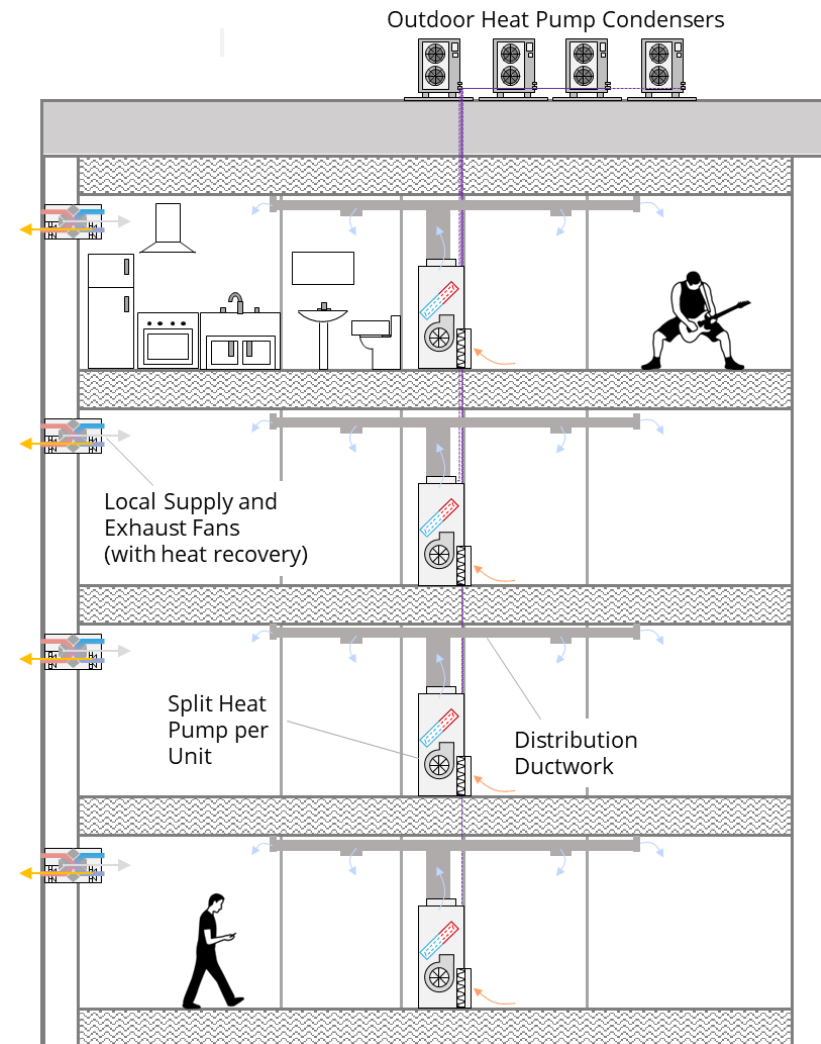
## Alternate 1d: MF Ventilation Option: Local Ventilator with Heat Recovery

System Tag: MF-ISUP\_IEXH-HRV-SZHP

### Mechanical Description

A single through-wall device per unit handles both supply and exhaust within a single housing, passing the two airstreams through a heat exchanger core. The exhaust air pre-conditions incoming supply air before it enters the unit, recovering a significant fraction of heating or cooling energy. This configuration

The Title 24 2025 standard system for this configuration is also individual balanced ventilation with a combined fan power of 0.60 W/cfm and a fixed heat recovery ventilator at SRE 0.67 / ASRE 0.72; the baseline HRV efficiency is fixed and does not adjust to match the efficiency of the proposed unit.



## Alternate 1e: MF Ventilation Option: Central Supply, Central Exhaust

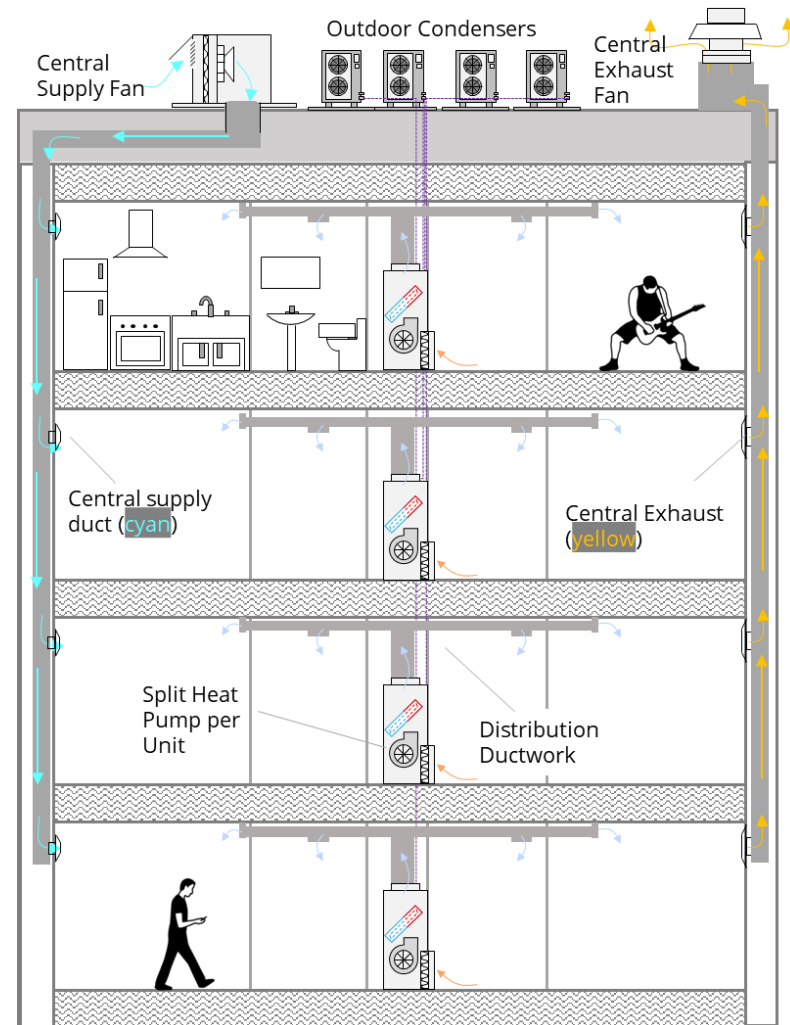
System Tag: MF-CSUP\_CEXH-HRV-SZHP

### Mechanical Description

#### (No Heat Recovery)

A single central fan delivers outdoor air to all units via a shared supply duct, and a separate central fan collects stale air from all units via a shared exhaust duct. Neither supply nor exhaust operates at the unit level. This configuration is more common in larger mid-rise buildings where centralizing both airstreams simplifies maintenance and allows for greater control over building pressurization.

The Title 24 2025 standard system for this configuration uses a central supply fan at 0.441 W/cfm and a central exhaust fan at 0.302 W/cfm, with no heat recovery ventilator in the baseline.



This example is one of eight developed as part of the *Addressing Complex HVAC Systems in Code Compliance* project. The full guide includes additional HVAC system types, alternative configurations, and working CBECC model files. All resources are available through the CalBEM working group and EnergyCodeAce platform.

